NICHOLAS YAP

Mechanical Engineer, EIT

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PROJECTS

Self-Balancing Wheeled Bipedal Robot: Prototype Grass – *Paired Project* **November 2025 – February 2026**

- Built a dual-PID self-balancing control system in collaboration with a partner to balance an elevated robot on two motorized wheels through accelerometer, gyroscope, and encoder input.
- Developed a C# WinForms application to interface with an ESP-WROOM-32 over BLE communication, allowing for wireless real-time PC control of the ESP through a client-server relationship.
- Designed and built a modular, multi-component chassis to integrate electrical systems, ensure precise sensor alignment, mitigate signal interference, and incorporate protective features to enhance safety and manufacturability.

Stepper Motor 2-Axis Gantry – *Paired Project* **September 2025 – December 2025**

- Soldered a predesigned PCB to implement a MSP430FR5739 microcontroller with motor drivers and encoder inputs.
- Programmed firmware in Code Composer Studio to control stepper motors for G-code–based gantry operation via UART from a C# WinForms application, enabling precise image drawing through custom G-code paths.

ENGINEERING EXPERIENCE

The Dow Chemical Company **February 2024 – August 2024**

Maintenance/Reliability Engineering Co-op Fort Saskatchewan, AB

- Reprogrammed an industrial oven PID controller and optimized equipment cleaning procedures, improving equipment reliability, maintenance efficiency, and operator safety.
- Designed SolidWorks flange covers for pressure equipment and coordinated Management of Change (MOC) processes to ensure safe implementation, regulatory compliance, and accurate technical documentation.
- Supported multidisciplinary maintenance projects by coordinating with operations, maintenance personnel, and equipment vendors to resolve technical issues, obtain engineering documentation, and implement equipment improvements.

The Dow Chemical Company **September 2023 – February 2024**

Fixed Asset Integrity Engineering Co-op Fort Saskatchewan, AB

- Developed and updated Corrosion Control Documents, Integrity Operating Windows, and ABSA regulatory documentation in collaboration with plant operations teams, supporting asset integrity, safe plant operation, and regulatory compliance.

UBC Uncrewed Aircraft Systems **September 2022 – Present**

Senior Aircraft and Administrative Member Vancouver, BC

- Designed airfoil frames, landing gear, and electrical mounts for autonomous VTOL and quadcopter aircraft using Onshape and SOLIDWORKS, collaborating with multidisciplinary teams to integrate mechanical, electrical, and software systems.

SKILLS

CAD & Design: SolidWorks, Onshape, Bluebeam

Programming & Analysis: Python, MATLAB, C, C++, C#, WinForms, PLC Ladder Logic

Instrumentation & Testing: Oscilloscope, Function Generator, Digital Multimeter, Power Supply

Embedded Systems: MSP430FR5739 Firmware, UART, BLE, PID Control, Code Composer Studio, Arduino, Soldering

EDUCATION

University of British Columbia **May 2026**

Bachelor of Applied Science in Mechanical Engineering, Mechatronics Specialization Vancouver, BC

CGPA: 87.6% – Dean's List | Graduated with Distinction